Field	Details
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Registration Number:	C11/26/PCDFIR/17017
Course Code	DF101
Course Tittle	Basic Computer Skills for Digital Forensics
Lab Tittle	Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis
Date	7 Nov 2026
Instructor	
Institution	International Cybersecurity and Digital Forensics Academy (ICDFA)

Ibrahim Zuleihat Eneze

[Company Name] | [Company Address]

Layer/Source	Example Evidence	Where it may be found	Why it matters
User Layer	Opened files,search items,typed usernames	Recent items,shell history	Links a user action to system activity
Application Layer	Chrome	Broswer profile folders	Shows what software created or used for the evidence
Os Layer	System logs,installed apps	Unified logs,/var/log	Shows system events and account activity
Disk layer	Deleted files,file fragments,partitions	File system structures	Support recovery and timeline analysis
Memory/process layer	Running malware,process IDs,network connections	RAM capture	Shows live system behaviour that disappers at shutdown

B1. Process Selection Table

Process Name	PID Value	Likely Purpose
/sbin/launchd	1	System Initialization:this is the very first process started by macOs during boot.
/usr/sbin/syslog	122	System Logging:this background daemon manages system logs
/usr/libexec/log	88	Unified Logging Engine:this core background process handless the high performance macOs unified logging system.

B2

Differences between a program,a process and a thread

Program ;is a file stored on a hard drive or storage device.
Process: is executing set of program loaded into the RAM.
Thread:is a unit of process.

Why process information is useful during malware analysis or live forensic triage
Detecting masquareading and anomalies
Tracing parent-child process relationship
Identifying hidden persistence and unallocated paths.

C2
Partition Name/ Identifier:disk0s2
Reported Size: 245.1GB
Number of Sectors (Sector Count):514,011,095
Bytes per Sector: 512
Total bytes:
514,011,955x512=263,174,120,960 bytes
Kilobytes(KB):
263,174,120,960\1024= 257,005,977.5KB
Megabytes (MB):
257,005,977.5/1024= 250,982.4MB

Gigabytes (GB):
250,982.4/1024=245.1GB

Partition Name/Identifier:disk0s3
Reported size:5.4GB
Number of Sectors (Sector Count):
11,324,620
Bytes per Sector:512

Total Bytes:
11,324,620x512=5,798,205,440 bytes
Kilobytes (KB):
5,798,205,440/1024=5,662,310 KB
Megabytes (MB):
5,662,310/1024=5,529.6MB
Gigabytes (GB):
5,529.6/1024=5.4GB

E1.The Boot Process Table & Explanation
My system uses UEFI.The evidence from my earlier diskutil list command shows a GUID_partition_scheme (GPT) and an Apple_APFS container layout,which is strictly paired with modern UEFI firmware on Apple hardware rather ythan legacy BIOS.

Boot Process Reference Table:

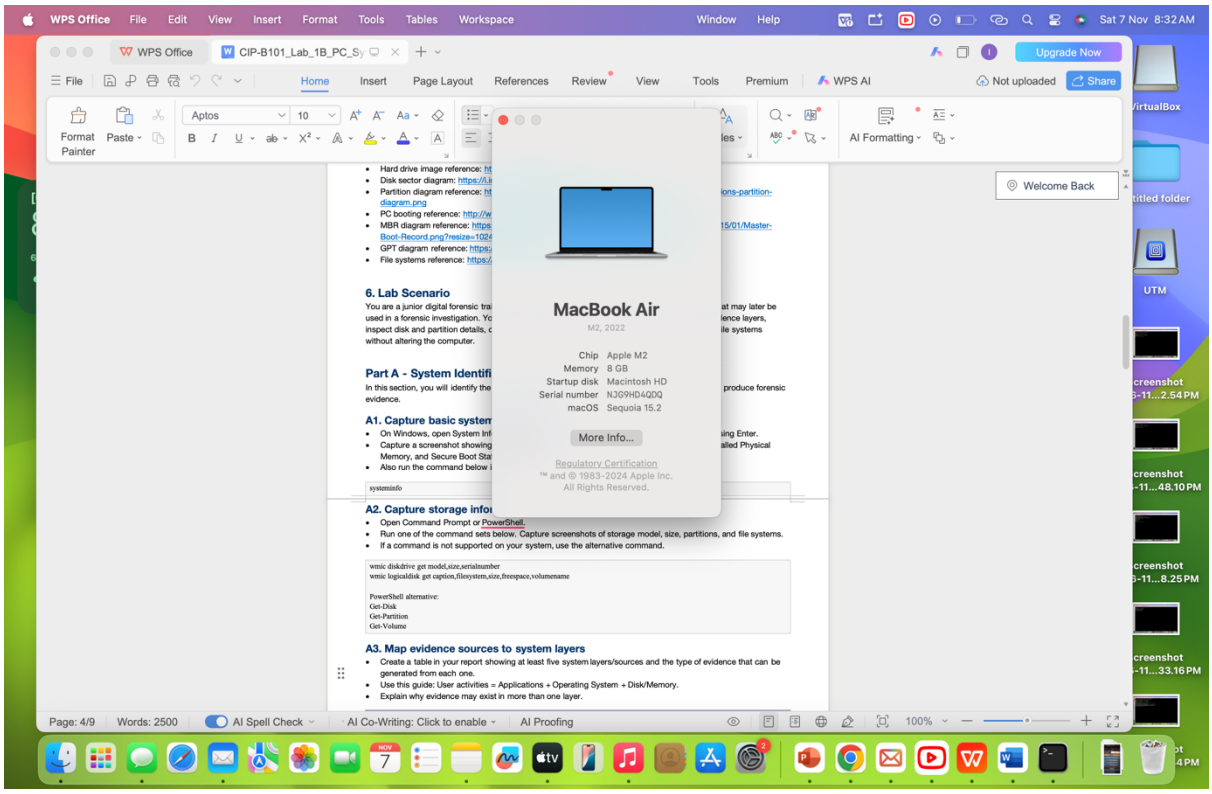
Stage	What Happens	Forensic Relevance
Power -on and POST	Electrical power stabilizes.The motherboard triggers the POST to check core hardware components like RAM,CPU,and internal buses.	Hardware failure unexpected hardware swaps,or physical tampering can affect access to evidence
BIOS/UEFI firmware	The firmware initializes from non-volatile memory,loads settings from CMOS,and applies the configured boot device order to locate the primary drive	Firmware setting can point to unauthorized external boot media (like a USB triage tool or malware booktits).
MBR/GPT	The firmware reads sector 0 for MBR layouts,or accesses the designated EFI system partition on a GPT drive to locate boot configuration execution flags.	Boot sector verification can reveal structural alterations,rootkits hidden outside regular file systems
Boot loader/Manager	The system hands off control to the specific Boot Loader	Helps detect hidden operating systems,multi boot modifications.
Kernel and OS control	The boot loader extracts the operating system kernel into memory	The exact handoff point where persistent operating system logs,volatile registry artifacts.

Feature	MBR	GPT
Age /Design	Older partitioning system introduced in the 1980s	Modern partitioning system used with contemporary UEFI platforms.
Maximum Disk Size	Limited to a maximum capacity of 2TB	Supports massive disk architectures up to 9.4 ZB
Primary partition limit	Restricted to a maximum of 4 primary partitions.	Allows up to 128 primary parttions natively under Windows/macOs
Redundancy & Safety	No backup tables;if the boot sector is corrupted,the drive layout is lost.	Keeps a duplicate Backup Partition Table at the very end of the disk for auto recovery
Integrity Checking	No built in vereification mechanism	Uses CRC32 to detect corruption in partition headers automatically.

G

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot 4	Disk0s2 (Apple_APFS- Macintosh HD)
Record partition 1 sector count	Screenshot 4	514,011,955
Calculate partition 1 sector size		Formula :sectors x 512/1024/1024/1024 Calculation:514,011,955x512=263,174,120,960 263,174,120,960/1.073,741,824= 245.1GB
Identify partition 2	Screenshot 4	Disk0s3 (Apple_APFS_Recovery-macOs Recovery)
Record partition 2 sector count	Screenshot 4	11,324,620 sectors
Calculate partition 2 size		Formula :sectors x 512/1024/1024/1024 Calculation : 11,324,620 x512=5,798,205,440 bytes 5,798,205,440/1,073,741,824 =5.4GB
Compare with OS-reprted size		The manual binary calculation match the sizes reported in the terminal command perfectly (245.1GB for partition 1 and 5.4GB for partition 2)
Explain difference if any		

SCREENSHOT ONE



SCREENSHOT TWO

```
      BSD Name: disk3s5
      Volume UUID: FD51DAC6-18C8-4B6E-A6AE-F40759DF8F4A
      Physical Drive:
          Device Name: APPLE SSD AP0256Z
          Media Name: AppleAPFSMedia
          Medium Type: SSD
          Protocol: Apple Fabric
          Internal: Yes
          Partition Map Type: Unknown
          S.M.A.R.T. Status: Verified

    VirtualBox:

      Free: 34.8 MB (34,840,576 bytes)
      Capacity: 194.7 MB (194,686,976 bytes)
      Mount Point: /Volumes/VirtualBox
      File System: APFS
      Writable: No
      Ignore Ownership: Yes
      BSD Name: disk7s1
      Volume UUID: DEEDC0EF-57B9-4B15-A08A-56A9B91552C3
      Physical Drive:
          Device Name: Disk Image
          Media Name: AppleAPFSMedia
          Protocol: Disk Image
          Internal: No
          Partition Map Type: Unknown

    Macintosh HD:

      Free: 146.02 GB (146,019,467,264 bytes)
      Capacity: 245.11 GB (245,107,195,904 bytes)
      Mount Point: /
      File System: APFS
      Writable: No
      Ignore Ownership: No
      BSD Name: disk3s1s1
      Volume UUID: CC13EFFF-33B3-4E61-9171-6A6460F4EABE
      Physical Drive:
          Device Name: APPLE SSD AP0256Z
          Media Name: AppleAPFSMedia
          Medium Type: SSD
          Protocol: Apple Fabric
          Internal: Yes
          Partition Map Type: Unknown
          S.M.A.R.T. Status: Verified

    UTM:

      Free: 580.6 MB (580,612,096 bytes)
      Capacity: 1.99 GB (1,988,059,136 bytes)
      Mount Point: /Volumes/UTM
      File System: HFS+
      Writable: No
      Ignore Ownership: Yes
      BSD Name: disk9s1
      Volume UUID: E67DA32C-0E86-330D-86E5-DAC98FF7D96D
      Physical Drive:
          Device Name: Disk Image
          Media Name: Apple Disk Image Media
          Protocol: Disk Image
          Internal: No
          Partition Map Type: GPT (GUID Partition Table)

mac@macs-MacBook-Air-2 ~ % PowerShell alternative:
```

SCREENSHOT THREE

```
mac@macs-MacBook-Air-2 ~ % [Insert Activity Monitor, or output of ps aux | head -20]
zsh: bad pattern: [Insert
head: illegal line count -- 20]
mac@macs-MacBook-Air-2 ~ % clear
```

```
mac@macs-MacBook-Air-2 ~ % ps -eo comm,pid,%cpu,%mem | head -n 21
COMM                PID  %CPU %MEM
/sbin/launchd         1   0.0  0.2
/usr/libexec/log     88   0.0  0.2
/usr/libexec/Use     90   0.0  0.1
/System/Library/     92   0.0  0.1
/System/Library/     93   0.0  0.1
/usr/sbin/system     96   0.0  0.1
/usr/libexec/con     99   0.0  0.1
/System/Library/   101   0.0  0.1
/usr/libexec/IOM   102   0.0  0.0
/usr/libexec/rem   106   0.0  0.0
/usr/libexec/wat   111   0.0  0.0
/System/Library/   115   0.0  0.2
/usr/libexec/ker   117   0.0  0.0
/usr/libexec/dis   118   0.0  0.0
/usr/sbin/syslog   122   0.0  0.0
/usr/libexec/the   125   0.0  0.0
/usr/libexec/ope   126   0.0  0.1
/System/Library/   128   0.0  0.1
/System/Library/   129   0.0  0.1
/usr/libexec/tim   130   0.0  0.0
mac@macs-MacBook-Air-2 ~ %
```

SCREENSHOT FOUR

```
mac@macs-MacBook-Air-2 ~ % diskutil list
/dev/disk0 (internal, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        *251.0 GB   disk0
   1:             Apple_APFS_ISC Container disk1         524.3 MB   disk0s1
   2:                 Apple_APFS Container disk3         245.1 GB   disk0s2
   3:        Apple_APFS_Recovery Container disk2         5.4 GB     disk0s3

/dev/disk3 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +245.1 GB   disk3
                                 Physical Store disk0s2
   1:                APFS Volume Macintosh HD            11.2 GB    disk3s1
   2:              APFS Snapshot com.apple.os.update-... 11.2 GB    disk3s1s1
   3:                APFS Volume Preboot                 6.8 GB     disk3s2
   4:                APFS Volume Recovery                1.0 GB     disk3s3
   5:                APFS Volume Macintosh HD - Data     77.6 GB    disk3s5
   6:                APFS Volume VM                      2.1 GB     disk3s6

/dev/disk4 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +194.7 MB   disk4
   1:                 Apple_APFS Container disk5         194.7 MB   disk4s1

/dev/disk5 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +194.7 MB   disk5
                                 Physical Store disk4s1
   1:                APFS Volume VirtualBox              158.4 MB   disk5s1

/dev/disk6 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +194.7 MB   disk6
   1:                 Apple_APFS Container disk7         194.7 MB   disk6s1

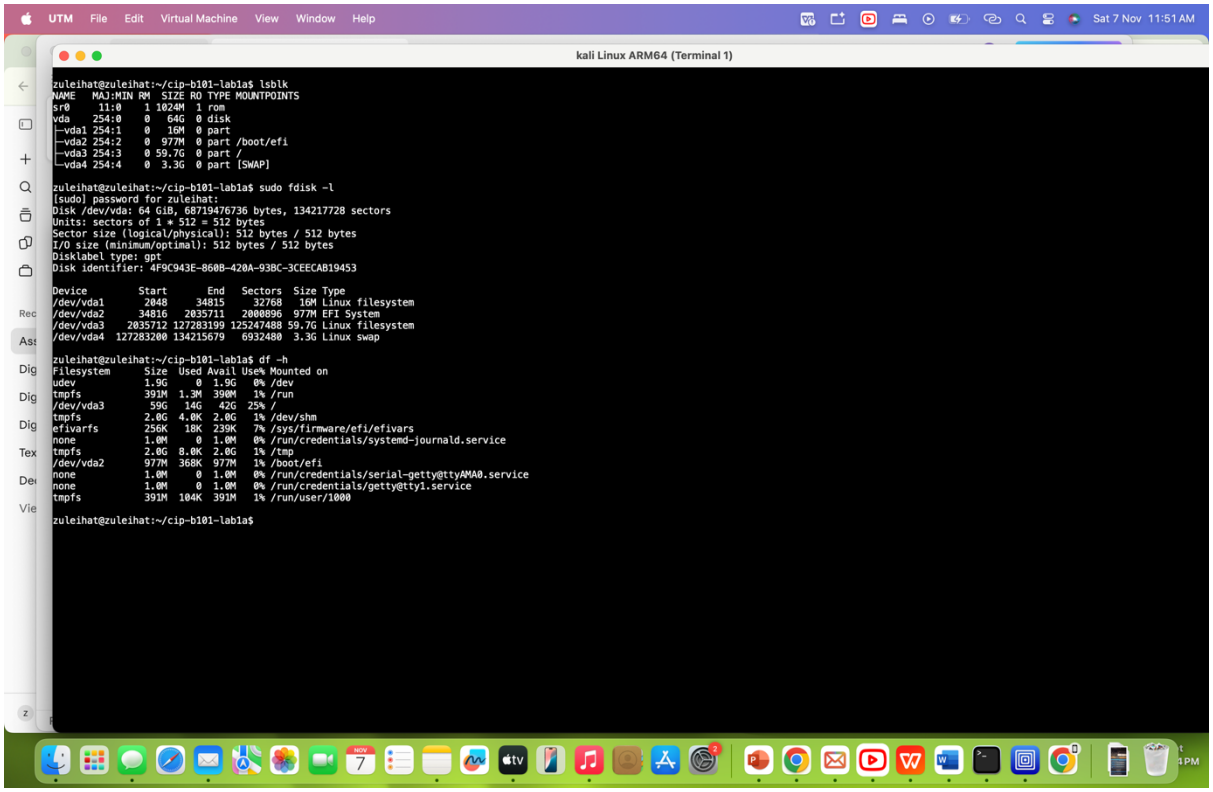
/dev/disk7 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +194.7 MB   disk7
                                 Physical Store disk6s1
   1:                APFS Volume VirtualBox              158.4 MB   disk7s1

/dev/disk8 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:     FDisk_partition_scheme                        +4.1 GB     disk8
   1:                       0xEF                         7.9 MB     disk8s2

/dev/disk9 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +2.1 GB     disk9
   1:                  Apple_HFS UTM                     2.0 GB     disk9s1

mac@macs-MacBook-Air-2 ~ %
```

SCREENSHOT FIVE



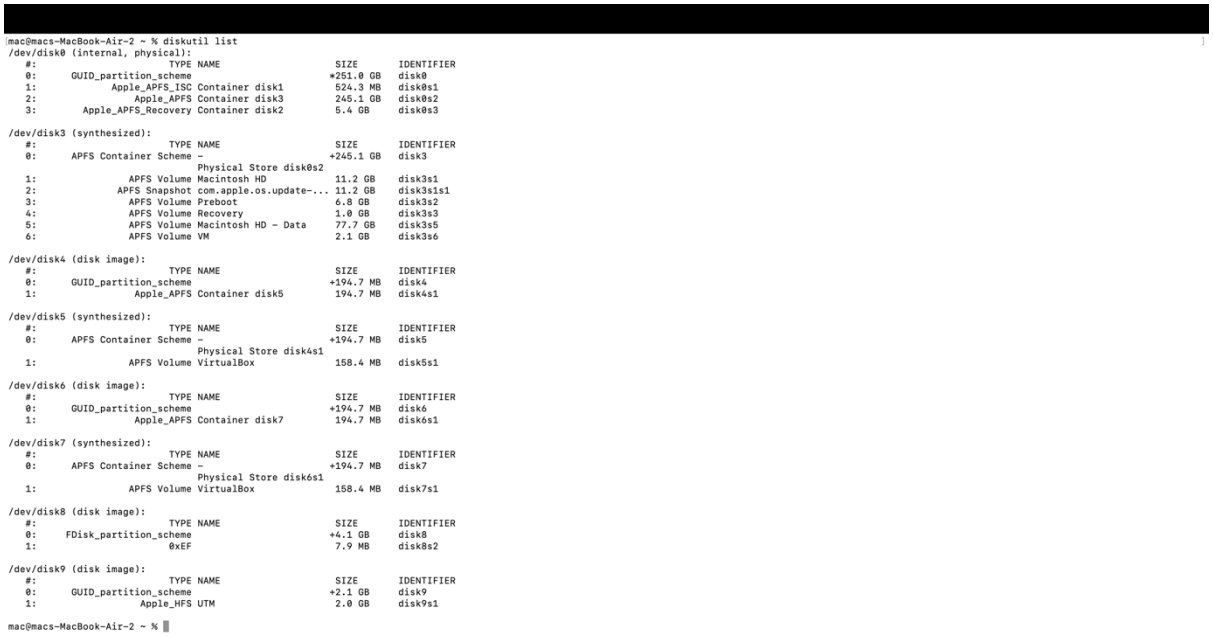
SCREENSHOT SIX



```
mac@macs-MacBook-Air-2 ~ % df -T
df: option requires an argument -- T
usage: df [-libxo] [-b | -g | -H | -h | -k | -m | -P] [-acIilnY] [-,] [-T type] [-t type]
          [file | filesystem ...]
mac@macs-MacBook-Air-2 ~ % clear
```

```
mac@macs-MacBook-Air-2 ~ % diskutil info disk0s2 | grep "Type (Bundle)"
mac@macs-MacBook-Air-2 ~ % mount | grep "on /"
/dev/disk3s1s1 on / (apfs, sealed, local, read-only, journaled)
devfs on /dev (devfs, local, nobrowse)
/dev/disk3s6 on /System/Volumes/VM (apfs, local, noexec, journaled, noatime, nobrowse)
/dev/disk3s2 on /System/Volumes/Preboot (apfs, local, journaled, nobrowse)
/dev/disk3s4 on /System/Volumes/Update (apfs, local, journaled, nobrowse)
/dev/disk1s2 on /System/Volumes/xarts (apfs, local, noexec, journaled, noatime, nobrowse)
/dev/disk1s1 on /System/Volumes/iSCPreboot (apfs, local, journaled, nobrowse)
/dev/disk1s3 on /System/Volumes/Hardware (apfs, local, journaled, nobrowse)
/dev/disk3s5 on /System/Volumes/Data (apfs, local, journaled, nobrowse, protect, root data)
map auto_home on /System/Volumes/Data/home (autofs, automounted, nobrowse)
/dev/disk3s3 on /Volumes/Recovery (apfs, local, journaled, nobrowse)
/dev/disk7s1 on /Volumes/VirtualBox (apfs, local, nodev, nosuid, read-only, journaled, noowners, quarantine, mounted by mac)
/dev/disk9s1 on /Volumes/UTM (hfs, local, nodev, nosuid, read-only, noowners, quarantine, mounted by mac)
mac@macs-MacBook-Air-2 ~ %
```

SCREENSHOT SEVEN



```
mac@macs-MacBook-Air-2 ~ % diskutil list
/dev/disk0 (internal, physical):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        *251.0 GB   disk0
   1:             Apple_APFS_ISC Container disk1         524.3 MB   disk0s1
   2:                 Apple_APFS Container disk3         245.1 GB   disk0s2
   3:       Apple_APFS_Recovery Container disk2         5.4 GB     disk0s3

/dev/disk3 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +245.1 GB   disk3
                                 Physical Store disk0s2
   1:                APFS Volume Macintosh HD            11.2 GB    disk3s1
   2:              APFS Snapshot com.apple.os.update-... 11.2 GB    disk3s1s1
   3:                APFS Volume Preboot                 6.8 GB     disk3s2
   4:                APFS Volume Recovery                1.0 GB     disk3s3
   5:                APFS Volume Macintosh HD - Data     77.7 GB    disk3s5
   6:                APFS Volume VM                      2.1 GB     disk3s6

/dev/disk4 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +194.7 MB   disk4
   1:                 Apple_APFS Container disk5         194.7 MB   disk4s1

/dev/disk5 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +194.7 MB   disk5
                                 Physical Store disk4s1
   1:                APFS Volume VirtualBox              158.4 MB   disk5s1

/dev/disk6 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +194.7 MB   disk6
   1:                 Apple_APFS Container disk7         194.7 MB   disk6s1

/dev/disk7 (synthesized):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      APFS Container Scheme -                      +194.7 MB   disk7
                                 Physical Store disk6s1
   1:                APFS Volume VirtualBox              158.4 MB   disk7s1

/dev/disk8 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:     FDisk_partition_scheme                        +4.1 GB     disk8
   1:                       0xEF                         7.9 MB     disk8s2

/dev/disk9 (disk image):
   #:                       TYPE NAME                    SIZE       IDENTIFIER
   0:      GUID_partition_scheme                        +2.1 GB     disk9
   1:                  Apple_HFS UTM                     2.0 GB     disk9s1

mac@macs-MacBook-Air-2 ~ %
```

Why is a file system important in digital forensics?
It tracks timestamps.
It finds deleted files.
It stores metadata.